Automation, factor shares and national income: an unanticipated AI shock in a UK overlapping-generations model*

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Abstract

I introduce artificial intelligence into OG-UK, the UK calibration of the OG-Core overlapping-generations model, as an unanticipated shock announced in 2028. AI lowers the labour share and raises output at given inputs over 2028–2031. Three scenarios are calibrated to UK sources (the IMF's 2026 UK Selected Issues paper and the OBR's productivity and fiscal-risks work), with the "substantial" US scenario of Korinek et al. (2026) as an upper case. Gaps are measured against an OBR-central path that already includes the OBR's own AI productivity assumption. In the UK central scenario, with OG-UK's default capital closure ($\zeta_K = 0.1$), gross national income in 2031 is 0.57 per cent above that path and GDP 0.92 per cent above it. The GNI gap is smaller because about three-quarters of the extra capital is foreign-owned and its return is paid abroad. Across the UK scenarios GNI is +0.3 to +2.0 per cent, and +3.8 per cent in the upper case. With private capital unable to flow in ($\zeta_K = 0$), the UK central GNI gap is +0.3 per cent. The real wage is below the path in every scenario in 2031, by 1.27 per cent in UK central, because the fall in the labour share outweighs the gain in output per hour. Aggregate consumption is also below the path in the UK scenarios while national income is above it, as households save out of a higher return. Two further arms bound the result from below: the same automation with technology unchanged, and an OBR-style stress test in which output at given inputs is set so that 2031 GDP is unchanged. In the stress test UK central GNI is −0.3 per cent. Debt/GDP is higher by 2036 in every arm, because the higher return reprices the whole stock of debt. The model's single income-tax schedule taxes labour about 14.8 percentage points more heavily than capital, against the OBR's 19–23 points, so it understates the fall in revenue.

Keywords: artificial intelligence, automation, overlapping generations, capital closure, national income, labour share
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1 Introduction

Quantified AI scenarios for the UK now come from several official and academic sources. The OBR's March 2026 *Economic and fiscal outlook* includes a technological-displacement scenario in which AI substitutes for labour, deepens capital and lowers the labour share (OBR, 2026a, Box 2.2). Its July 2026 *Fiscal risks and sustainability* report sets out a range for the AI boost to productivity growth and an illustrative fall in the wage share (OBR, 2026b, Box 4.1). The IMF's 2026 UK Selected Issues paper models AI adoption across sectors with an endogenous capital share (IMF, 2026). Korinek et al. (2026) quantify a "substantial" US scenario with a larger labour-share fall and productivity gain. These sources specify factor-share and technology shocks. None works out how a UK economy with a calibrated tax system, an ageing population and an open capital account would respond to them in general equilibrium, or whose income the response raises.

This paper does that in OG-UK, the UK calibration of the OG-Core overlapping-generations model (DeBacker and Evans, 2024). The model has UN demographic projections, eighty annual cohorts with endogenous saving and labour supply, and effective tax functions estimated from PolicyEngine UK microdata. AI enters production

$$Y_t = Z_t \, K_t^{\gamma_t} \left(e^{g_y t} L_t\right)^{1-\gamma_t} \tag{1}$$

through the capital exponent γ_t, which carries automation, and the technology term Z_t. The economy follows an OBR-central path to the end of 2027. At the start of 2028 the whole AI path is announced, unanticipated, and γ and Z then move over 2028–2031. The headline year is 2031, the first year at full strength.

The scenarios are imposed, not estimated. I calibrate three to UK sources: UK low, UK central and UK high, with labour-share falls of 0.7, 1.6 and 2.9 percentage points and output gains at given inputs of 0.2, 0.4 and 1.6 per cent by 2031 (Section 3.4). The "substantial" scenario of Korinek et al. (2026), a 3.9-point fall with a 3.1 per cent gain, is kept as a labelled upper case. Each scenario is run in three arms: "automation + productivity", the main result; "automation, technology unchanged"; and an "OBR-style stress test: no output gain", in which output at given inputs is set so that 2031 GDP equals its path. All gaps are measured against the OBR-central path, which already includes the OBR's own AI productivity assumption, so they are AI effects beyond that assumption, not against a world without AI.

The paper makes four points.

National income rises by less than GDP. How much of the extra capital is supplied from abroad depends on the capital closure. Under OG-UK's default ($\zeta_K = 0.1$, with $\zeta_D = 0.4$ and a 2 per cent world interest rate), 77 per cent of the extra capital in UK central is foreign-owned in 2031, and its return is paid abroad. I therefore lead with gross national income (GNI) and also report net national income (NNI). In UK central, GNI in 2031 is +0.6 per cent against the path, NNI +0.3 per cent and GDP +0.9 per cent. With the capital account closed to private capital ($\zeta_K = 0$) the GNI gap is +0.3 per cent. Both closures are reported together throughout (Table 2).

The size depends on the scenario, and the UK range is well below the US upper case. Across the three UK scenarios, 2031 GNI is +0.3, +0.6 and +2.0 per cent at the central closure; in the upper case it is +3.8 per cent. For the same productivity input, UK central's

GDP gain is about a third of the IMF's output gain at the same horizon. The difference is the capital response, which in this model is set by the closure and by household saving that has had three years to build (Section 4.5).

Wages and consumption fall while national income rises. At a unit elasticity of substitution the wage is $(1-\gamma_t)Y_t/L_t$, so it rises only if output per hour rises by more than the labour share falls. In every scenario and arm it does not by 2031: in UK central the real wage is 1.27 per cent below the path at the central closure and 1.71 per cent below it at the bound. Aggregate consumption is also below the path in the UK scenarios in 2031 (-0.39 per cent in UK central), because households save more out of the higher return on capital and government consumption rises with GDP.

The public finances worsen by 2036 in every arm. Debt/GDP is below the path in the main arm in 2031 at the central closure (-0.13 points in UK central), but it is above it by 2036 in every arm, scenario and closure ($+1.99$ points in UK central), because the higher interest rate reprices the whole stock of debt. Revenue falls as a share of GDP at the central closure, but by less than the OBR's effective tax rates imply, because OG-UK taxes labour and capital income through one income-tax schedule (Section 4.4).

The model's cohort structure also allows incidence by birth cohort and lifetime-earnings group. Section 5 sets out the method. Its results wait on the debt-neutral runs, in which transfers adjust so that households, not the public debt, bear the fiscal cost of the shock.

The model has no unemployment, no task reinstatement and no adoption friction, and capital has no adjustment cost. Section 8 sets out these and the other boundaries. The rest of the paper reviews the literature (Section 2), sets out the model, the timing, the scenarios and the closure (Section 3), reports the results (Section 4) and the incidence method (Section 5), and discusses them (Section 6). Appendix A.4 documents the patch that lets OG-Core's capital exponent change over time.

2 Related literature

Automation in growth models. The mechanism simulated here is the oldest one in the automation literature. Zeira (1998) shows that when technology moves tasks from labour to capital, output rises while labour's share of it falls, and the aggregate capital exponent rises with the automated share of tasks. Acemoglu and Restrepo (2018) add a countervailing force, separating the displacement effect of automation from the reinstatement effect of new labour-intensive tasks. Acemoglu and Restrepo (2022) find that task displacement accounts for a substantial part of the post-1980 rise in US wage inequality. In that framework the exponent is an equilibrium object. Here it is a parameter, which is the cost of running the mechanism inside a calibrated fiscal overlapping-generations model.

Automation, wealth and factor shares. Moll et al. (2022) show that automation redistributes towards wealth: capital income is more concentrated than labour income, so a shock that raises the return on wealth and lowers the labour share raises inequality through portfolios as well as wages. Acemoglu (2025) argues that task-level exposure limits the attainable aggregate gain. Jones and Tonetti (2026) calibrate a task-based growth model with an elasticity of

substitution across tasks of 0.2 and find that such "weak links" make the growth effect of automation slow. Moll and Imas (2026), in an essay, raise the capital exponent of a Cobb-Douglas economy with a constant saving rate from one third to one by 2045, and cite that low elasticity as a reason why growth would be slower than such a model implies. The elasticities in these papers are across tasks; Korinek et al. (2026) set theirs to 0.5 and the elasticity of capital supply to 3. The elasticity fixed at one here is between capital and labour.

Automation in overlapping-generations models. Sachs et al. (2015) build an overlapping-generations model of automation and find that a rise in robot productivity is more likely to lower the welfare of young workers and future generations when the saving rate is low, and that transfers across generations can leave every generation with a gain. This paper extends that question to UK-calibrated AI scenarios, in a model calibrated to UK data with an estimated tax system, and sets out incidence by birth cohort and lifetime-earnings type together rather than young against old.

Quantified AI scenarios. Korinek et al. (2026) supply the upper case used here. Their framework has an automation share, an affected task mass, an adoption parameter and a re-instatement parameter that build from a mid-2026 anchor to 2030, and a matching block that produces equilibrium unemployment. The "substantial" column of their Table 3 reports, for the US in 2030, a 3.9 percentage-point fall in the labour share, measured TFP 3.1 per cent above the no-AI path, GDP 8.3 per cent above it, a capital stock 13.8 per cent higher and unemployment of 4.6 per cent. Four of their six AI channels have no counterpart in OG-UK (Appendix A.5). IMF (2026) model AI adoption in the UK with 100 sectors and an endogenous capital share that rises by about 10 percentage points in the long run; TFP is 0.53 per cent and output 3.26 per cent above a no-AI path five years after adoption starts. Output exceeds TFP there because capital deepens strongly.

The UK official benchmarks. In the OBR's March 2026 technological-displacement scenario, capital deepens, the level of GDP is unchanged against the central forecast, higher average labour productivity does not pass through to real earnings, the labour share falls and the profit share rises. Equilibrium unemployment rises to 5.5 per cent, and receipts fall because labour income faces a higher effective tax rate than capital income (OBR, 2026a, Box 2.2 and para 6.18). The OBR's central forecast already includes an AI productivity assumption, built up in Annex B of OBR (2025b): a boost to productivity growth of about 0.2 percentage points a year by the five-year horizon, with alternatives from below 0.1 to 0.8 points (OBR, 2026b, Box 4.1). In the same box the OBR shows that a fall in the wage share lowers receipts, because labour income faces a higher effective tax rate than profits. The OBR's own overlapping-generations model (OBR, 2025a) is close in architecture to OG-UK. It runs either closed or as a small open economy, and has not been used for an AI scenario.

Capital mobility. Whether extra investment is financed at home or abroad is an old empirical question. Feldstein and Horioka (1980) found that national saving and investment moved almost one for one across OECD countries in 1960–74. Blanchard and Giavazzi (2002) find much lower coefficients for the European Union and the euro area in 1991–2001. Section 3.6 uses these estimates to judge the model's capital closure.

The modelling platform. OG-Core (DeBacker and Evans, 2024) is an open-source descendant of the Auerbach–Kotlikoff framework (Auerbach and Kotlikoff, 1987). OG-UK is its UK calibration, with demographics from UN projections and tax functions from PolicyEngine UK.

3 Model, timing, scenarios and closure

3.1 The model

OG-UK is the UK calibration of OG-Core (DeBacker and Evans, 2024), an open-source overlapping-generations model in the Auerbach–Kotlikoff tradition (Auerbach and Kotlikoff, 1987). Households live for eighty annual periods from age 21. They are divided into seven lifetime-earnings types with an age- and type-specific productivity profile, and they choose consumption, saving and hours. Firms are competitive and produce with equation (1), which at the elasticity of substitution used here, $\varepsilon = 1$, is Cobb-Douglas. Government levies taxes through effective tax functions estimated from PolicyEngine UK microdata, spends a share of output, pays transfers and issues debt. Fertility, mortality and immigration come from UN projections. Technology grows at the labour-augmenting rate g_y. All runs are single-sector.

At $\varepsilon = 1$ the labour share is exactly $1 - \gamma_t$, and the real wage satisfies

$$w_t = (1 - \gamma_t)\,\frac{Y_t}{L_t}, \qquad \Delta \ln w_t = \Delta \ln(1 - \gamma_t) + \Delta \ln \frac{Y_t}{L_t}. \tag{2}$$

The labour-share path is therefore an input, not a result. Factor quantities, prices, national income and the public finances are results.

3.2 Timing: an unanticipated shock in 2028

Every run follows the same path in 2026 and 2027: the OBR-central baseline described below. At the start of 2028 the whole future path of γ and Z is announced. It was not anticipated, so nothing in 2026–2027 responds to it. From 2028 households and firms know the path and re-plan from the wealth they hold. γ and Z move with weights 0, 0.25, 0.5, 0.75 and 1 on their terminal values over 2027–2031 and stay there afterwards, so 2028–2031 are the first four years of adoption.

OG-Core starts every run from the baseline's initial state, so a run cannot simply begin in 2028. Each AI run is built from the same 2026 specification as the baseline, with every time-indexed input shifted forward two years, the pre-period demographic objects set to the baseline's 2027 values, and the initial state replaced by the baseline's state at the start of 2028: the wealth each cohort carries into that year, the bequests it implies, government debt and its foreign-held share, and public capital. Nothing in OG-Core or OG-UK is edited for this. After each solve, the run's 2028 household wealth, debt and domestically owned capital must equal the baseline's to 10^{-6}, or the result is not used. The fiscal rule still starts in 2036.

All gaps are measured against a no-shock restart built the same way, with γ and Z at their baseline values. The restart reproduces the committed baseline over 2028–2036 to a relative error of at most 5.1×10^{-6} in GDP and 2.3×10^{-5} in labour input, consumption, the return, the wage, debt, household wealth and investment (as a share of GDP) at the central closure (1.2×10^{-6} and 8.2×10^{-6} at the bound). Measuring against the restart removes the solver's own noise from every gap. I call it "the path" below.

Three features of this timing matter for reading the results.

- *2028 is the announcement-year response plus the first quarter of the change.* On the news, households expect lower wages and higher returns, so they work and save more at once; and γ and Z take their first step that year.

- *Investment jumps on the news.* OG-Core 0.17 has no adjustment cost on capital, so investment is whatever moves the capital stock to next year's target. The 2028–2030 investment profile is a property of frictionless capital, not a forecast of an investment boom. I report the 2031 levels.

- *At $\zeta_K = 0.1$ capital moves on impact.* Only household wealth, debt and domestically owned capital carry over from 2027. Foreign-owned capital is reset every year by the closure rule (equation 3), so part of the 2028 capital gap is an inflow from abroad with no investment recorded in 2027. At $\zeta_K = 0$ this does not happen.

3.3 The baseline

The baseline sets $\gamma = 0.35$ (a labour share of 65 per cent), $Z = 1$ and $g_y = 1.0$ per cent a year, the OBR's medium-term productivity growth rate. That rate already contains the OBR's own AI assumption. Briefing paper No. 9 puts the boost to productivity growth at about 0.2 percentage points a year at the five-year horizon, between a J-curve path that reaches 0.1 points and an S-curve path that reaches 0.25 points (paras B.12 and B.14), with a range from below 0.1 to 0.8 points (OBR, 2025b, 2026b). A scalar g_y cannot carry that path, so the baseline keeps the OBR's rate and is labelled "OBR central". Every gap in this paper is therefore relative to the OBR-central path, which already includes the OBR's own AI assumption; none is a comparison with a world without AI. Section 8 discusses the overlap this creates with the productivity targets.

The switch of government spending to debt targeting is set at $t_{G1} = 10$ (2036). The baseline's spending share before the switch is set so that its 2030 debt-to-GDP ratio matches the OBR's March 2026 forecast of 95.1 per cent. The baseline ratio is 95.2 per cent in 2031 and 96.9 per cent in 2036. Appendix Table 5 compares the baseline's growth with the OBR's forecast.

3.4 The scenarios

Mapping. At $\varepsilon = 1$ a labour-share fall of f points maps to the capital exponent as $\gamma_1 = 0.35 + f/100$. Raising γ alone changes output at given inputs, and that change reflects the units of K and L, not technology. So Z is solved jointly with γ to hit a target for output at given inputs: the baseline's 2027 capital and labour, the last year every run shares. The solved Z lies below one in every arm only because it offsets the γ-only change; its level has no economic meaning.

Targets. Table 1 gives the four scenarios. All targets are levels reached in 2031, the fourth year of adoption.

- *UK central.* The two UK central estimates agree at year four. The IMF's adoption path gives TFP about 0.42 per cent above no AI at year four (IMF, 2026, Table 1 and Figure 9), and a linear ramp to the OBR's 0.2 points a year by year five gives 0.4 per cent. The IMF's model ties the labour-share fall to the TFP gain through a 27 per cent cost saving per automated task (IMF, 2026, para 44), which gives about 1.6 points at year four (0.42/0.27,

my inference from their calibration; a straight line to their long-run 10 points would give 2.35 points but overstates the early years of a convex adoption path).

- *UK low.* The OBR's J-curve path, 0.1 points a year by year five, gives 0.2 per cent at year four; the same IMF mapping gives a 0.7-point share fall.
- *UK high.* The OBR's high alternative of up to 0.8 points a year gives 1.6 per cent at year four. For the share, the OBR's "reduced labour share" illustration, which it places at the upper end of the literature, has the wage and salary share falling linearly from 40 per cent of GDP in 2030–31 to 20 per cent in 2075–76 (OBR, 2026b, Box 4.1). Four years of that path is 1.78 points of GDP, which is 2.9 points when scaled to the model's 65 per cent labour share (my mapping). The IMF mapping would give 5.9 points here, above the OBR's own upper end, so the share fall is capped at the OBR path.
- *Upper case: Korinek et al., US "substantial".* The labour share falls from 60.0 to 56.1 per cent and measured TFP is 3.1 per cent higher in 2030 (Korinek et al., 2026, Table 3). This is a US scenario. Its TFP gain is about eight times UK central's and its share fall about 2.4 times. It is labelled as an upper case throughout, not as a UK calibration.

The OBR figures are for labour-productivity growth, which includes capital deepening, so used as TFP they overstate it somewhat; the IMF figure is TFP proper and agrees at the central value. Because the target is set at 2027 inputs and the path's capital per hour falls after 2027, the realised gain at 2031 inputs is 2–6 per cent smaller than each target.

Table 1: The scenarios: targets reached in 2031, and sources

Scenario	Labour-share fall, pp	γ in 2031	Output at given inputs, %	Sources
UK low	0.7	0.357	0.2	OBR BP9 J-curve (para B.12); IMF task mapping
UK central	1.6	0.366	0.4	IMF SIP/2026/072, year 4 (Table 1, Figure 9, para 44); OBR central 0.2pp a year (BP9 para B.14; FRS 2026 Box 4.1)
UK high	2.9	0.379	1.6	OBR FRS 2026 Box 4.1: 0.8pp a year alternative; reduced-labour-share path, scaled to a 65% share
Korinek et al. (upper)	3.9	0.389	3.1	Korinek et al. (2026, Table 3), US "substantial", 2030

Notes: the baseline has $\gamma = 0.35$ and a 65 per cent labour share. Output at given inputs is measured at the baseline's 2027 capital and labour. Year-4 values from published paths are my arithmetic: a linear ramp to g points a year by year five gives a level of $2g$ per cent at year four. Derivations: `docs/uk_calibration.md`; targets: `analysis/surprise/uk_targets.json`.

3.5 Three arms

Each scenario is run three ways. All three have the same labour-share path.

- *"automation + productivity"* (the main result). Z hits the scenario's output target at given inputs.

- *"automation, technology unchanged".* Z is set so that output at the baseline's 2027 inputs is unchanged: automation reallocates income from labour to capital with no gain in output at given inputs. No UK source describes this case; it isolates the share shift.

- *"OBR-style stress test: no output gain".* Z is solved, at each closure, so that GDP in 2031 equals the path to within 0.01 percentage points. This follows the OBR's displacement scenario, in which the level of GDP is unchanged (OBR, 2026a, Box 2.2), but it is not a reproduction of it. It is a pessimistic bound, not a forecast: real AI adoption implies some productivity gain (Acemoglu, 2025; IMF, 2026). GDP is held in 2031 only and drifts afterwards.

Showing the stress test beside "automation, technology unchanged" makes clear that its zero is imposed. At the central closure, holding GDP flat needs output at given inputs to fall, by 0.42 per cent against "automation, technology unchanged" in UK central, because the capital inflow alone raises GDP. At the closed-capital bound it needs a small rise instead (0.03 per cent in UK central and 0.05 per cent in the upper case), because without a foreign inflow "automation, technology unchanged" already leaves GDP slightly below the path. There the two arms nearly coincide.

Every arm uses the baseline's government spending share, $\alpha_G = 0.299$, and transfer share, so government consumption and transfers are the same share of output in every arm until the debt rule starts in 2036. Two variants change that convention and are being solved: fixed spending, which holds government consumption and transfers at the path's levels, as the OBR does when scoring a scenario against a forecast; and debt-neutral transfers, in which transfers adjust every year from 2028 so that debt/GDP follows the path to 2036.

3.6 The capital closure

OG-Core's open-economy block sets the foreign-owned capital stock as a fixed share ζ_K of the gap between the capital firms would demand at the world interest rate, K_t^{open}, and the domestically owned stock K_t^d:

$$K_t = K_t^d + \zeta_K\big(K_t^{\text{open}} - K_t^d\big). \tag{3}$$

Foreigners also buy a share ζ_D of new government debt. Gross national income is GDP less the return paid on foreign-owned capital and foreign-held debt, $\text{GNI}_t = Y_t - r_t^p(K_t^f + D_t^f)$. Net national income is $\text{NNI}_t = \text{GNI}_t - \delta K_t$ with $\delta = 0.065$, OG-UK's annual depreciation rate; it is 81.3 per cent of GDP on the path in 2031.

The central closure is OG-UK's default: $\zeta_K = 0.1$, $\zeta_D = 0.4$ and a world interest rate of 2 per cent. **This is an inherited default, not a UK calibration.** OG-Core's documentation sets $\zeta_K = 0.1$ for a large open version of the US economy and notes that it is hard to pin down. The case for keeping it is after the fact, and rests on two comparisons with UK data:

- *Income paid abroad.* The path pays 1.58 per cent of GDP abroad in 2031. The UK's net primary income deficit was 1.2 per cent of GDP in 2025 and 1.6 per cent in 2024 (ONS, 2026). This is the quantity that separates GNI from GDP.

- *Who finances extra investment.* In UK central, extra national saving covers 0.34 of the extra investment over 2028–2031. That is a model analogue of the saving–investment coefficient of Feldstein and Horioka (1980), and it lies close to the estimate of Blanchard and Giavazzi (2002, Table 3) for the European Union in 1991–2001 (0.36).

The closure gets other features wrong:

- The path's current-account deficit is 1.1 per cent of GDP in 2031, against the UK's 3.0 per cent in 2025 (ONS, 2026). Foreigners own 8 per cent of the capital stock, against about 40 per cent gross in the OBR's estimate (OBR, 2025a).
- The size of the inflow depends on the 2 per cent world rate, which no run has varied. The path's domestic return is 5.3 per cent, so firms' demand for capital at the world rate far exceeds the actual stock, and equation (3) acts on that standing gap.

The closed-capital bound sets $\zeta_K = 0$, with ζ_D and the world rate unchanged. It is not a closed economy, since foreigners still buy 0.4 of new gilts. It matches the UK on neither comparison above: the path pays 0.56 per cent of GDP abroad, and extra national saving covers 0.82 of extra investment, close to the 0.89 that Feldstein and Horioka (1980) found for 1960–74. I report it beside the central case because it bounds the role of foreign capital. Each closure has its own baseline and restart, and gaps are measured against them.

3.7 Solution and reporting

Runs used OG-Core 0.17.0 and OG-UK at commit d0e6ae5, with PolicyEngine UK 2.90.2. Every committed solve passes a sanity gate: the accounting identities, the resource constraint, equality with the baseline in 2026–2027, and the labour-share path. Appendix A.2 reports the errors. Every number in the paper is a macro generated by `paper/make_values.py` from the committed per-solve records, which also checks every result the text states in words.

All results are gaps between a run and the path in the same year, at the same closure. Gaps are reported for 2031. Debt/GDP is also reported for 2036, the first year of the debt rule. No other series is quoted for 2036, because in that year the rule cuts government consumption sharply in every run and the flows are not comparable; series that go beyond 2031 stop at 2035.

4 Results

All results are gaps against the path (the no-shock restart) at the same closure, in 2031 unless stated. The central closure is $\zeta_K = 0.1$ with $\zeta_D = 0.4$ and a 2 per cent world rate; the bound is $\zeta_K = 0$. The path is OBR central and already includes the OBR's own AI assumption.

4.1 The headline: UK central

Table 2 is the paper's main result. In "automation + productivity", GNI in 2031 is +0.6 per cent against the path at the central closure and +0.3 per cent at the bound. GDP is +0.9 and +0.4 per cent. NNI is +0.3 and +0.2 per cent.

GNI moves less with the closure than GDP does, because at the central closure 77 per cent of the extra capital is foreign-owned and most of the extra output it produces is paid to its owners. NNI moves less still: it also deducts depreciation on the extra capital, which is located in the UK whoever owns it. A GDP gap overstates what UK residents receive whenever foreign capital responds, so I lead with GNI.

Table 2: UK central in 2031: three arms at the central closure and the closed-capital bound

	"automation + productivity"		Technology unchanged		Stress test	
	$\zeta_K = 0.1$	$\zeta_K = 0$	$\zeta_K = 0.1$	$\zeta_K = 0$	$\zeta_K = 0.1$	$\zeta_K = 0$
GNI, %	+0.57	+0.30	+0.15	−0.10	−0.30	−0.06
NNI, %	+0.27	+0.23	−0.21	−0.24	−0.72	−0.20
GDP, %	+0.92	+0.37	+0.47	−0.03	0.00	0.00
Capital stock, %	+2.01	+0.67	+1.88	+0.63	+1.73	+0.63
Labour input, %	−0.30	−0.40	−0.29	−0.38	−0.27	−0.38
Real wage, %	−1.27	−1.71	−1.71	−2.12	−2.19	−2.09
Consumption, %	−0.39	−0.52	−0.61	−0.73	−0.84	−0.71
Investment, %	+3.58	+3.31	+2.78	+2.45	+1.94	+2.52
Return on capital, bp	+38	+48	+34	+44	+31	+44
Labour share, pp	−1.60	−1.60	−1.60	−1.60	−1.60	−1.60
Tax/GDP, pp	−0.19	−0.01	−0.21	−0.05	−0.24	−0.04
Debt/GDP, pp, 2031	−0.13	+0.33	+0.35	+0.78	+0.87	+0.74
Debt/GDP, pp, 2036	+1.99	+1.98	+2.67	+2.67	+3.40	+2.61

Notes: gaps against the no-shock restart at the same closure ("the path"), which is OBR central and includes the OBR's own AI assumption. UK central: labour share −1.6 points and output at given inputs +0.4 per cent by 2031. Technology unchanged: the same share fall with output at given inputs unchanged. Stress test: output at given inputs set so that 2031 GDP equals the path. GNI is $Y - r^p(K^f + D^f)$; NNI is GNI $-\delta K$. Debt/GDP is quoted with its 2036 value, the debt rule's first year. Source: `analysis/surprise/output/`.

Where the extra capital comes from. The shock raises the marginal product of capital, and capital rises in response in every arm at both closures. The closure decides who supplies it. At the central closure about three-quarters of the extra capital in 2031 is foreign-owned, the capital response is larger and the rise in the return smaller. At the bound, household saving supplies all of it: capital rises 0.67 per cent against 2.01 per cent, and the return rises 48 basis points against 38. The capital-supply elasticity (the change in log capital over the change in the log rental rate) is 0.59 at the central closure in 2031 and 0.16 at the bound, rising to 1.39 and 1.06 in the steady state. Capital is a stock built from saving, and by 2031 household saving has had three years (2028–2030) to respond, while the return moved at once.

The announcement year and the profile. In 2028 investment, hours and saving all rise in every arm at both closures, and consumption falls: in UK central's main arm at the central closure investment is +7.82 per cent, labour input +0.17 per cent and consumption −0.55 per cent against the path. Households learn that wages will be lower and returns higher, so they work and save more at once. Labour input then turns down as the wage falls and wealth builds. Table 3 gives the year-by-year path. Investment stays high while γ and Z are still rising and drops in 2031, when they stop; with no adjustment cost, that profile is a property of the model, not a forecast.

4.2 Wages, consumption and national income

The labour share follows the imposed path, falling 1.6 points in UK central by 2031. At $\varepsilon = 1$ that is the input restated. The wage follows equation (2). In UK central the share fall alone takes 2.5 log points off the wage. Output per hour gives part of it back, through the technology

Table 3: UK central, "automation + productivity", central closure: gaps against the path by year

	2027	2028	2029	2030	2031	2035	2036
GNI, %	0.00	+0.28	+0.39	+0.49	+0.57	+0.57	
GDP, %	0.00	+0.36	+0.56	+0.74	+0.92	+0.94	
Capital, %	0.00	+0.43	+0.95	+1.49	+2.01	+2.31	
Labour input, %	0.00	+0.17	+0.03	−0.13	−0.30	−0.43	
Real wage, %	0.00	−0.42	−0.71	−0.99	−1.27	−1.12	
Consumption, %	0.00	−0.55	−0.52	−0.47	−0.39	−0.09	
Investment, %	0.00	+7.82	+8.41	+8.90	+3.58	+2.84	
Return, bp	0	+12	+21	+29	+38	+35	
Debt/GDP, pp	0.00	−0.34	−0.37	−0.30	−0.13	+1.57	+1.99

Notes: 2027 is before the announcement, so every gap is zero by construction. 2028 is the announcement-year response plus the first quarter of the change in γ and Z. Only debt/GDP is shown for 2036, the debt rule's first year, when government consumption is cut in every run. Source: `analysis/surprise/output/zk0.1/`.

gain and through capital deepening, but not all: the real wage is −1.27 per cent at the central closure and −1.71 per cent at the bound in 2031, and it is still below the path in 2035. The wage is below the path in every scenario and arm in 2031.

Across scenarios the wage does not follow the size of the shock. At the central closure it is −0.53, −1.27 and −1.27 per cent in UK low, central and high, and −0.59 per cent in the upper case. It depends on the ratio of the productivity gain to the share fall, which is lowest in UK central. Per unit of GDP gained, UK central is the least favourable of the four for wages.

Consumption falls while national income rises. In UK central's main arm at the central closure, GNI is above the path and aggregate consumption below it in every year from 2028 to 2035 (−0.39 per cent in 2031). The same holds in 2031 for UK low and UK high, and at the bound. Two uses take the extra income and more. Government consumption is a fixed share of GDP, so it rises with GDP and takes part of the GNI gain. And households save more: labour income falls, household capital income rises with the higher return, and households build the wealth that the larger capital stock and the extra public debt require. Only in the upper case at the central closure, where the productivity gain is large, is consumption above the path by 2031 (+0.28 per cent).

4.3 The range across scenarios

Table 4 reports the main arm across the four scenarios. At the central closure GDP, GNI, capital and the return all rise with the size of the shock: GNI is +0.27, +0.57 and +2.02 per cent in the three UK scenarios and +3.80 per cent in the upper case. GDP is close to additive in the two parts of the shock. The share shift with technology unchanged raises GDP at the central closure, through the capital inflow, and the productivity gain adds to that.

The stress test. With GDP held at the path in 2031, GNI is below the path in every scenario at the central closure (−0.30 per cent in UK central) and the wage falls further than in the main arm (−2.19 per cent). GNI falls because part of the unchanged output is now paid abroad as the return on foreign-owned capital. At the bound GNI is −0.06 per cent in UK central. GDP is held in 2031 only. Afterwards it drifts below the path at the central closure (−0.08 per cent

Table 4: "automation + productivity" across scenarios, 2031

	UK low	UK central	UK high	Korinek et al. (upper)
Labour-share fall, pp (input)	0.7	1.6	2.9	3.9
Output at given inputs, % (input)	0.2	0.4	1.6	3.1
Central closure, $\zeta_K = 0.1$				
GNI, %	+0.27	+0.57	+2.02	+3.80
NNI, %	+0.14	+0.27	+1.58	+3.33
GDP, %	+0.41	+0.92	+2.74	+4.89
Capital, %	+0.87	+2.01	+4.09	+6.03
Real wage, %	−0.53	−1.27	−1.27	−0.59
Consumption, %	−0.16	−0.39	−0.20	+0.28
Return, bp	+17	+38	+76	+109
Tax/GDP, pp	−0.08	−0.19	−0.29	−0.35
Debt/GDP, pp, 2031	−0.07	−0.13	−1.35	−3.02
Debt/GDP, pp, 2036	+0.84	+1.99	+2.07	+1.12
Closed-capital bound, $\zeta_K = 0$				
GNI, %	pending	+0.30	pending	+2.88
GDP, %	pending	+0.37	pending	+3.05
Real wage, %	pending	−1.71	pending	−2.03
Tax/GDP, pp	pending	−0.01	pending	+0.21
Debt/GDP, pp, 2036	pending	+1.98	pending	+1.03
"OBR-style stress test: no output gain", central closure				
GNI, %	−0.13	−0.30	−0.57	−0.79
Real wage, %	−0.95	−2.19	−3.98	−5.36
GDP, %, 2035	−0.03	−0.08	−0.17	−0.23

Notes: gaps against the path at the same closure. "Pending" marks solves still queued. The upper case is a US scenario, not a UK calibration. The stress test holds GDP in 2031 only; its 2035 row shows the drift afterwards. Source: `analysis/surprise/output/`.

in UK central by 2035), as debt rises and crowds out domestically owned capital, and above it at the bound (+0.07 per cent), where capital keeps building from household saving. The stress test is a bound on the downside, not a central estimate.

4.4 The public finances

Revenue. At the central closure tax revenue falls as a share of GDP in every scenario, arm and year from 2028 to 2035: by 0.19 points in UK central's main arm and 0.24 points in its stress test in 2031. That is the direction the OBR describes, but the size is smaller. Per point of labour-share fall, the stress test lowers tax/GDP by 0.15 points in UK central (0.15–0.16 across scenarios), against 0.19–0.23 in the OBR's illustration, which also holds GDP unchanged (OBR, 2026b, Box 4.1).

The reason is the tax system in OG-UK. Employee NICs are inside the estimated income-tax function, which is fitted on PolicyEngine UK income tax plus employee NICs and applies one schedule to total household income, labour and capital alike. Employer NICs are a flat 15 per cent payroll tax on all labour income. On the path in 2031 the effective rate on labour is 34.4 per cent and on gross capital income 19.6 per cent, a gap of 14.8 points (13.5 at the bound). The OBR's rates imply a gap of 19–23 points (46 against 27 per cent in the fiscal-risks report, 40 against 17 in the November 2025 outlook; OBR, 2026b, 2025c). So each point of income moved from labour to capital lowers revenue by less in the model than in the OBR's arithmetic. Two further differences point the same way: the business tax rises with the capital share, and the progressive income-tax schedule takes more as the return on household wealth rises.

At the closed-capital bound the second channel dominates. With no foreign inflow the return rises more, on all household wealth including gilts. In the upper case's main arm tax/GDP *rises*, by 0.21 points in 2031 and in every year from 2029: the payroll tax falls by 0.59 points and consumption taxes by 0.30, but the income-tax function takes 0.41 points more, as household capital income rises by 4.50 points of GDP, and the business tax 0.71 more. In UK central's main arm at the bound tax/GDP is −0.01 points in 2031 and above zero from 2032. Every OBR source has revenue falling when the labour share falls, so these bound results are the wrong sign. They are a known limitation of the single income-tax schedule (Section 8). Fiscal results at the bound should be read with that in mind.

Debt. Debt/GDP is higher by 2036 in every arm, scenario and closure: by 1.99, 2.67 and 3.40 points in UK central's three arms at the central closure. In the main arm at the central closure the 2031 gap is negative (−0.13 points in UK central), because higher GDP lowers the ratio while the gap is opening, but it turns positive in 2032 and rises after that. A 2031 figure alone would therefore give the wrong sign of the path, and I quote debt with its 2036 value throughout. The cause is the interest rate. The model's government debt reprices in full each year at a rate that moves with the return on capital, so the whole stock, about 95 per cent of GDP, pays more from the first year. Under the shared-share convention used here, government consumption and transfers rise with GDP, so higher output does not pay the extra interest back; tax/GDP falls as well. UK gilts have long maturities, so the actual stock would reprice over many years, and the model overstates the speed of this effect.

The fixed-spending and debt-neutral variants are solved overnight. Appendix Table 7 reports them when they are complete: [pending: fixed-spending and debt-neutral solves, both closures].

4.5 Against published scenarios

The IMF. For the same productivity input, UK central's main arm gives GDP +0.92 per cent at year four at the central closure, against the IMF's 2.5 per cent at year four and 3.3 per cent at year five (IMF, 2026); about a third. The IMF's figures are measured against no AI, and this paper's against the OBR-central path, which already includes the OBR's own AI assumption. The difference is the capital response. If capital were supplied at an unchanged return, the rental rate $\gamma Y/K$ would stay at its path value, and with UK central's inputs that needs capital 7.9 per cent higher and gives GDP 3.2 per cent above the path with labour unchanged (2.9 per cent with the arm's own labour). That brackets the IMF's figures. In the solve the return rises 38 basis points instead, and capital rises 2.01 per cent. The calculation shows that the closure and the horizon are sufficient to account for the difference; it does not show that the IMF model works the same way.

Korinek et al. The upper case reproduces the labour-share fall and productivity gain of their "substantial" scenario. GDP rises 4.89 per cent at the central closure and 3.05 per cent at the bound, against their 8.3 per cent, and capital 6.03 and 1.80 per cent against their 13.8 per cent. Their own Table 5 shows the same dependence on capital supply: their average wage is −1.6, +2.1, +3.7 and +5.6 per cent against their no-AI path at capital-supply elasticities of 1, 3, 6 and infinity. In this model the elasticity in 2031 is below 1 at both closures, and the upper case's wage is −0.59 and −2.03 per cent.

The OBR. The model reproduces three qualitative features of the OBR's displacement scenario at the central closure: the labour share falls, the profit share rises and revenue falls as a share of GDP. It cannot reproduce the rise in equilibrium unemployment. The stress test reproduces the OBR's unchanged level of GDP by construction, and its GNI row shows what that implies for national income at each closure. UK low is the closest in size to the OBR's displacement case.

5 Incidence across cohorts

The aggregate results hide who gains and who loses. The model has eighty annual cohorts and seven lifetime-earnings groups, so the same runs give incidence by birth cohort and by group, which representative-agent and two-group AI models cannot report. This section sets out the method. The results wait on the debt-neutral runs and are marked pending.

Why the debt-neutral runs. In the main runs the shock changes public borrowing, and no tax or transfer responds before the debt rule starts in 2036; from then the rule adjusts government consumption, which does not enter utility. A welfare measure computed on those runs would count any extra borrowing that reaches households, through lower taxes or higher interest on the gilts they hold, as a gain that no cohort repays. The headline cohort results therefore use the debt-neutral variant: from 2028 transfers adjust every year so that debt/GDP follows the path to 2036, and households bear the fiscal effect of the shock through transfers instead of through public debt. The variant is solved at both closures for UK central and the upper case. [pending: debt-neutral solves (UK central and Korinek, both closures)]

Measure. Household utility in OG-Core has three parts: constant-relative-risk-aversion utility of consumption, an elliptical utility of leisure, and a warm-glow utility of the assets left as a bequest, scaled by the probability of dying. For each birth cohort b and lifetime-earnings group j, I compute remaining lifetime utility on the path and in each scenario from the household paths of the solves themselves: consumption, labour supply and assets by year, age and group. Future periods are discounted by the discount factor, the survival probabilities and the growth adjustment $e^{g_y(1-\sigma)}$ that OG-Core's detrended problem implies; the adjustment is the same in every run. The welfare change $\lambda_{b,j}$ is the constant percentage change in the path's consumption and bequests, in every remaining year, that gives the household the scenario's remaining lifetime utility, with labour supply held at its path. Cohorts alive in 2028 are measured over their remaining lives from 2028, when the shock is announced, and later cohorts over their whole lives. Each closure is measured against its own path.

Three checks tie the measure to the aggregates: the household paths, weighted by the population and group shares, must reproduce the records' consumption gaps in every year; the path compared with itself must give zero; and the path with consumption and bequests 1 per cent higher must give exactly 1 per cent. The solved transition ends before the economy reaches its steady state, so later years are filled with each run's last age profile, and the results are also reported with each run's steady state instead.

The groups are percentiles of lifetime earnings: 0–25, 25–50, 50–70, 70–80, 80–90, 90–99 and the top 1 per cent. They are not income deciles; where a comparison with household income is useful, I refer to the bottom and top income deciles explicitly.

How the change is taken. The measure counts consumption, leisure and bequests together. Each result will also be reported with one of the three parts at a time switched from the path to the scenario, so that a reader who values consumption alone can use the consumption term. The split describes the form in which households take the change, not its cause, because labour supply and saving are chosen.

What the measure depends on. Four features of the model will shape these results, and each is reported with the size of its effect where the runs allow.

- *Transfers.* Lump-sum transfers are the same amount per head. In the debt-neutral runs they also carry the fiscal effect of the shock, so their path depends on the scenario.
- *Wealth.* Household wealth in the model is spread more evenly across lifetime-earnings groups than earnings are, because every group has the same discount factor and bequests are shared within groups. A higher return on capital therefore reaches lower lifetime earners at least as much as higher ones. UK wealth is far more concentrated: the top tenth of households by wealth held 43 per cent of household wealth in 2018–20 and the bottom half 9 per cent (ONS, 2022). In Moll et al. (2022) the concentration of capital income is a main channel from automation to inequality, and that channel is weak here.
- *Foresight from the announcement.* Households learn the whole path in 2028 and re-plan at once. A model in which the change is learned gradually would leave the cohorts alive in 2028 less time to adjust.
- *Leisure and bequests are valued.* Labour supply falls by 2031, and the measure counts the extra leisure as a gain. The bequest term values the assets a household carries forward,

which rise with the return.

Results. [pending: welfare change by birth cohort and lifetime-earnings group, debt-neutral runs, UK central and Korinek, both closures; figure and parts table]

6 Discussion

6.1 What the model adds to the scenarios

The scenarios specify a labour-share fall and a productivity gain. At a unit elasticity the model turns these into the labour-share path mechanically. Its contribution lies elsewhere. It shows how much capital the shocks draw in, who owns that capital, how much of the extra output stays with UK residents, and what happens to wages, consumption and the public finances on the way. Any statement that AI raises UK output by a given amount carries an assumption about how capital is financed: in UK central that assumption moves 2031 GDP from +0.4 to +0.9 per cent, while GNI moves from +0.3 to +0.6 per cent and NNI hardly at all (+0.2 and +0.3 per cent).

6.2 What is robust and what is not

- *Robust across scenarios, arms and closures (where solved).* Capital and the return on capital rise. The real wage is below the path in 2031. Debt/GDP is higher by 2036. In 2028 hours and saving rise and consumption falls.
- *Robust across closures, not across scenarios.* In the UK scenarios aggregate consumption is below the path in 2031 while GNI is above it; in the upper case at the central closure consumption is above the path.
- *Not robust to the closure.* The size of the GDP and GNI gains; the sign of the change in tax/GDP in the main arm (it falls at the central closure and rises at the bound once the productivity gain is large enough); the sign of GDP in "automation, technology unchanged" (above the path at the central closure, slightly below it at the bound); and whether the stress test needs output at given inputs to fall or to rise slightly.
- *Not robust to the year.* The sign of the debt/GDP gap in the main arm at the central closure, negative in 2031 and positive from 2032.
- *Not yet tested.* The fiscal convention (fixed spending and debt-neutral transfers, solved overnight); the 2 per cent world interest rate and ζ_D; an elasticity of substitution below one; separate tax schedules for labour and capital income; and gradual learning of the shock.

6.3 Reading the fiscal result

At the central closure revenue falls as a share of GDP in every scenario and arm, which is the direction the OBR describes, but by less than the OBR's effective tax rates imply. The shortfall comes from the model's tax system, which taxes labour about 14.8 points more than capital against the OBR's 19–23, and whose single income-tax schedule takes more when the return on household wealth rises. The fall would follow from any shock that moved income from labour

to capital by the same amount, so it is a property of the UK tax system as much as of AI. Debt rises by 2036 for a separate reason: the higher return reprices the debt stock, and the shared-share spending convention spends the growth dividend.

7 Conclusion

I introduce AI into a UK overlapping-generations model as an unanticipated shock announced in 2028, calibrated to UK sources, and report its general-equilibrium effects on capital, labour, national income and the public finances relative to the OBR-central path, which already includes the OBR's own AI assumption. In the UK central scenario, with OG-UK's default capital closure, GNI in 2031 is +0.6 per cent above that path and GDP +0.9 per cent; across the UK scenarios GNI is +0.3 to +2.0 per cent, and +3.8 per cent in the US upper case. With the capital account closed to private capital the UK central GNI gain is +0.3 per cent. The default closure was inherited, not calibrated to the UK, and the after-the-fact case for it rests on the UK's income paid abroad and on published saving–investment estimates. The real wage and, in the UK scenarios, aggregate consumption are below the path in 2031 while national income is above it. Debt/GDP is higher by 2036 in every arm. The OBR-style stress test, in which output at given inputs is set so that GDP is unchanged in 2031, gives a GNI gap of −0.3 per cent in UK central and is a pessimistic bound, not a forecast. The natural next steps are the debt-neutral and fixed-spending runs, which are under way, cohort incidence on the debt-neutral runs, separate labour and capital tax schedules in OG-UK, an elasticity of substitution below one, and a world interest rate set to the path's domestic return.

8 Limitations

The baseline already contains an AI assumption, and UK central partly draws on it. The path uses the OBR's medium-term productivity growth of 1.0 per cent a year, which includes the OBR's central AI boost of about 0.2 percentage points a year by the five-year horizon (OBR, 2025b). Every gap in this paper is therefore relative to the OBR-central path, not to a world without AI. UK central's productivity target (+0.4 per cent by 2031) is supported by two sources: the IMF's TFP path, which is measured against no AI, and a linear ramp of the same OBR central assumption. To the extent that UK central is read as the OBR's AI path, it is counted twice: once in the baseline's growth rate and once in the scenario. Two treatments would remove the overlap, and the choice between them is open. (a) Keep the baseline and describe UK central as AI productivity in addition to the OBR's central path, with its target resting on the IMF's estimate. (b) Rebase the path to productivity growth without the OBR's AI assumption (about 0.8 per cent a year) and measure every scenario against it, which requires re-solving the baselines and every scenario. Until that choice is made, the numbers should be read as effects beyond the OBR's central AI assumption.

The tax system has one income-tax schedule. OG-UK fits a single effective income-tax function, on income tax plus employee NICs from PolicyEngine UK, to total household income, and levies employer NICs as a flat 15 per cent payroll tax on all labour income. The effective rate on labour is 34.4 per cent and on gross capital income 19.6 per cent on the path, a gap of 14.8 points against the OBR's 19–23. Two consequences follow. Revenue falls by too little when income moves from labour to capital, and at the closed-capital bound tax/GDP rises in

the upper case's main arm (+0.21 points in 2031) and in UK central's main arm from 2032, the opposite of every OBR source. And the flat payroll tax is above the average employer rate: PolicyEngine UK 2.90.2 gives employer NICs of about 11 per cent of total labour cost in 2028 (12.3 per cent of employment income), because of the secondary threshold and the Employment Allowance, against the model's flat 15 per cent. Lowering it would narrow the labour–capital gap further. The fix is separate labour and capital tax functions, which OG-Core supports and OG-UK does not yet estimate; it is outside this paper.

The capital closure is inherited. $\zeta_K = 0.1$ is OG-UK's default, carried over from OG-Core's US documentation. The case for it here is after the fact (Section 3.6). Its current-account deficit is 1.1 per cent of GDP against the UK's 3.0 per cent. The inflow depends on a 2 per cent world interest rate that no run has varied, and ζ_D was not varied either.

Capital has no adjustment cost. OG-Core 0.17 defines an adjustment-cost function but does not use it. Investment is whatever moves capital to next year's target, so it jumps on the announcement and drops when the ramp ends. The year-by-year investment profile is not a forecast. At the central closure part of the 2028 capital gap arrives from abroad with no investment recorded the year before.

Government debt reprices at once. The model's debt is one-period, and its rate moves with the return on capital, so the whole stock pays the higher rate from the first year. UK gilts have long maturities, so the model overstates how fast higher rates raise debt service, and with it the 2036 debt gaps.

The spending convention. In the reported runs government consumption and transfers are fixed shares of GDP until the debt rule starts in 2036, so higher output raises spending. The fixed-spending variant, the OBR's convention, and the debt-neutral variant are being solved. From 2036 the debt rule cuts government consumption in every run, so flows in that year are not reported.

The labour share is an input. At $\varepsilon = 1$ the labour share is exactly $1 - \gamma_t$. The scenario's fall is an assumption carried through the model, not evidence for it. With an elasticity below one, capital deepening would move factor shares, and the mapping from automation to output and wages would change. No run varies ε; OG-Core indexes it by industry only, at about 20 sites that would need the treatment Appendix A.4 gives γ.

The scenario targets are partly inferred. The UK sources give productivity paths, not labour-share paths at year four. The UK low and central share falls are inferred from the IMF's cost-saving calibration, and the UK high fall maps the OBR's wage-and-salary share onto the model's factor share. The OBR productivity figures include capital deepening. The upper case is a US scenario.

No unemployment, reinstatement or adoption friction. The model has no labour-market block, so the OBR's rise in equilibrium unemployment and the 4.6 per cent of Korinek et al. (2026) have no counterpart. Labour input changes through hours chosen at a different wage, not through job loss. There is no creation of new labour-intensive tasks (Acemoglu

and Restrepo, 2018), so the labour share cannot recover, and no adoption friction beyond the imposed ramp.

Foresight from the announcement. The shock is unanticipated before 2028, and from 2028 households know its whole path. A model in which the change is learned gradually would show a smaller announcement-year response.

The solved Z levels have no economic meaning. They lie below one because they offset the change in output at given inputs that moving γ alone produces, which reflects the units of K and L. Only the joint move of γ and Z is interpretable.

One sector, one calibration. All runs are single-sector and deterministic. There is no uncertainty quantification over γ, Z, ε, the demographic projection or the tax functions; the scenario range is the only sensitivity to the size of the shock.

The patch is not upstream. The time-varying capital exponent (Appendix A.4) is applied to the installed OG-Core 0.17.0. Results cannot be reproduced from a stock installation without it.

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Appendix

A.1 The baseline against the OBR's March 2026 forecast

The baseline takes its productivity growth rate from the *Economic and fiscal outlook* and is then compared with it. Its debt/GDP is calibrated so that 2030 is matched (Section 3.3); it is not compared in the table.

Table 5: The baseline ($\zeta_K = 0.1$) against the OBR, March 2026 EFO

	OBR	This model	Gap, points
Real GDP growth, 2027–2030 mean, per cent	1.6	1.57	−0.03
Productivity growth, medium term, per cent	1.0	1.01	
Baseline growth, 2027		1.61	
Baseline growth, 2028		1.54	
Baseline growth, 2029		1.58	
Baseline growth, 2030		1.57	

Notes: OBR figures from the March 2026 EFO, paragraphs 1.2, 1.9 and 2.10. The productivity rate includes the OBR's own AI assumption. Source: `uk_growth_with_ai/data/scenarios.json` and `uk_growth_with_ai/obr.py`.

A.2 Solution diagnostics

Every solve is checked after it runs (`analysis/surprise/sanity.py`): the accounting identities $K = K^d + K^f$, $K^d = B - D^d$ and $D = D^d + D^f$; the resource constraint; equality with the baseline in 2026 and 2027; the 2028 stocks against the baseline's; and the labour-share gap against its target in each year of the ramp. Table 6 gives the largest errors across all the solves used in the paper. The no-shock restart's agreement with the committed baseline is in Section 3.2.

Table 6: Solution diagnostics, all solves used

	$\zeta_K = 0.1$	$\zeta_K = 0$
Solves (restart and arms)	14	7
Max resource-constraint error, share of GDP, after 2028	4.1×10^{-6}	2.1×10^{-6}
Max Euler error	5.6×10^{-12}	5.0×10^{-12}
Longest solve, minutes	8.7	7.2

Notes: OG-Core's `ENFORCE_SOLUTION_CHECKS` is off, so solution quality is recorded per solve rather than enforced at solve time. Investment follows from the capital law of motion; it is not a residual. Source: `analysis/surprise/output/`.

A.3 Fixed spending and debt-neutral transfers

Table 7 will report UK central and the upper case under two alternative fiscal conventions: government consumption and transfers held at the path's levels, the convention the OBR uses when scoring a scenario against a forecast; and transfers that adjust every year from 2028 so that debt/GDP follows the path to 2036. In each, the stress test's output at given inputs is re-solved so that 2031 GDP again equals the path.

Table 7: UK central and the upper case under alternative fiscal conventions, "automation + productivity", 2031

	UK central			Korinek et al. (upper)		
	Shared share	Fixed spending	Debt-neutral	Shared share	Fixed spending	Debt-neutral
Central closure, $\zeta_K = 0.1$						
GNI, %	+0.57	pending	+0.68	+3.80	pending	pending
GDP, %	+0.92	pending	+1.02	+4.89	pending	pending
Consumption, %	−0.39	pending	−0.56	+0.28	pending	pending
Debt/GDP, pp, 2036	+1.99	pending	0.00	+1.12	pending	pending
Closed-capital bound, $\zeta_K = 0$						
GNI, %	+0.30	pending	pending	+2.88	pending	pending
GDP, %	+0.37	pending	pending	+3.05	pending	pending
Consumption, %	−0.52	pending	pending	−0.16	pending	pending
Debt/GDP, pp, 2036	+1.98	pending	pending	+1.03	pending	pending

Notes: gaps against the path at the same closure. "Shared share" is the convention of the main text. "Pending" marks solves queued overnight. In the debt-neutral runs debt/GDP follows the path by construction to 2036. Source: `analysis/surprise/output/`.

A.4 A time-varying capital exponent for OG-Core

Stock OG-Core indexes the capital exponent by industry only, as `p.gamma[m]`, at about 14 sites in `ogcore/firm.py`. Automation can therefore step to a new level in the first period but cannot phase in. The scenarios used here phase in over 2028–2031, and Moll and Imas (2026) raise the exponent over two decades. Neither can be run without a time dimension.

The patch (`patches/ogcore-0.17.0-firm-gamma-tv.diff`, against OG-Core 0.17.0) adds one in a backward-compatible way. A helper indexes a production parameter that may carry either shape, following the convention Z already uses: the last row in the steady state, the first T rows along the transition.

```
def _tv(x, p, method, m=None):
    x = np.asarray(x)
    if x.ndim <= 1:                 # (M,) -- time-invariant, legacy
        return x[m] if m is not None else x
    if method == "SS":              # (T, M) -- time-varying
        return x[-1, m] if m is not None else x[-1]
    return x[: p.T, m] if m is not None else x[: p.T]
```

Every `p.gamma[m]`, `p.gamma_g[m]` and bare `p.gamma` is replaced by `_tv(p.gamma, p, method, m)`. A one-dimensional parameter is returned exactly as before, so existing calibrations are bit-identical. `get_MPx` also needs the share in column shape along the transition, or a $(T,)$ share forms a (T, T) outer product.

As a regression check, the single-sector scenario with a one-dimensional exponent reproduces the unpatched run exactly. As a capability check, an exponent ramping from 0.35 to 0.389 solves and produces a labour-share path rather than a step. The runner refuses to start against an unpatched `ogcore.firm`. The patch is not upstream.

Table 8: How the AI channels of the comparison models map onto this one

Channel	Korinek et al. (2026)	Moll and Imas (2026)	This model
Automation	Automation share; affected task mass ramps	Capital exponent $1/3 \to 1$	γ, phased in (Appendix A.4)
Augmentation	The non-automated share	None	Absorbed into Z
Adoption and diffusion	Explicit adoption parameter	—	None
Task reinstatement	Explicit reinstatement parameter	—	None
Unemployment and search	Matching block	—	None
Capital accumulation	Capital-supply elasticity	Solow saving rate	Cohort saving plus foreign capital (ζ_K)
Labour share	Endogenous	$\to 0$ in the limit	Exactly $1-\gamma_t$ at $\varepsilon = 1$
Worker heterogeneity	Two groups	Representative agent	Seven earnings types, eighty ages
Fiscal system	None	None	Tax functions from PolicyEngine UK
Ideas and R&D	Ideas stock	Discussed, not in the baseline	None

Notes: Z absorbs capability, adoption and productivity together, because the model has no tasks to count separately, and γ carries automation. Reinstatement and search have no counterpart.

A.5 Mapping the AI channels across models

A.6 Reproduction

Runs used OG-Core 0.17.0 with the patch above, OG-UK at commit d0e6ae5 and PolicyEngine UK 2.90.2. OG-UK reads PolicyEngine's UK data from Hugging Face, so a re-run needs a token. The design and the solve plan are in `docs/surprise_2028.md`; the calibration in `docs/uk_calibration.md`. The paper reads only committed files. From the repository root:

```
python paper/make_values.py --check     # values_generated.tex is current
cd paper && tectonic main.tex           # or: pdflatex main && pdflatex main
```

Every number is a macro in `paper/values_generated.tex`, generated from the per-solve records in `analysis/surprise/output/<closure>/`. `make_values.py` stops with an error if a record is missing (other than the solves still queued, which print as pending), if a record fails its checks, or if a sentence that the paper states in words (for example, that debt/GDP is higher by 2036 in every arm) no longer holds.